Freight Rate Prediction

Objective:

To predict freight rate for every load using Machine learning model to support daily operations of the company.

Data:

Three sets of data are given as:

1. Train-Test data (Contains columns: load_id, pickup, delivery, pickup_lat, pickup_lon, delivery_lat, delivery_lon, distance, equipment, weight, date, market_index, quote_signal and posted_rate).
2. Validation data (Contains all the columns mentioned in train-test data except the posting_rate).
3. December chart inputs (contains columns: pickup, delivery, distance, equipment and date. Except for the date column, values for the remaining columns are fixed).

Data Analysis:

Data analysis is divided into following sections.

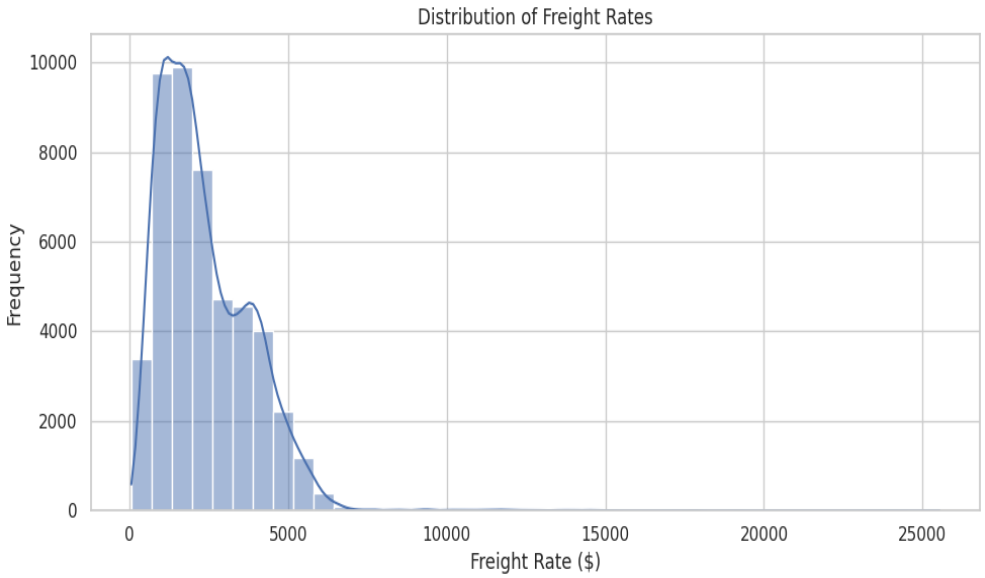
1. **Data processing**:
    - Train-test data contained 48000 rows and 14 columns.
    - Validation data consisted 12000 rows and 13 columns.
    - Negative values from the “weight” column in both train-test and validation data were replaced by missing values.
    - “weight” and “market_index” columns of both train-test and validation data contained missing values. These values were replaced by the median values of respective columns.
    - Outliers in both train-test and validation data were capped and floored with the quantile values.
    - “Date” column in train-test and validation data was converted to data type “datetime”.

2. **Feature engineering**:
   Following features were derived from the data.

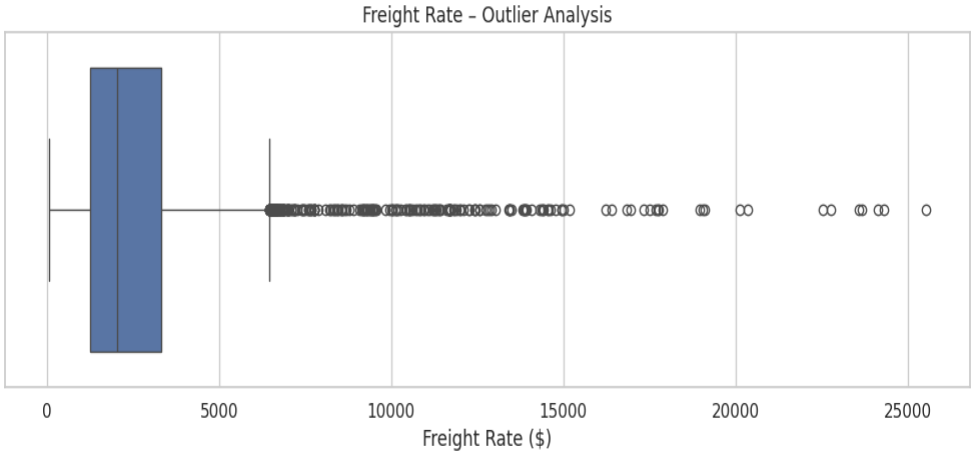
Feature	Description
abs_latitude_diff	Absolute latitude difference
abs_longitude_diff	Absolute longitude difference
geographic_distance_miles	Straight-line distance between pickup and delivery
reported_to_geographic_distance_ratio	Reported distance / Geographic distance
distance_per_weight_miles_per_lb	Distance / Weight
weight_per_mile_lb_per_mile	Weight / Distance

3. **Exploratory Data Analysis**:

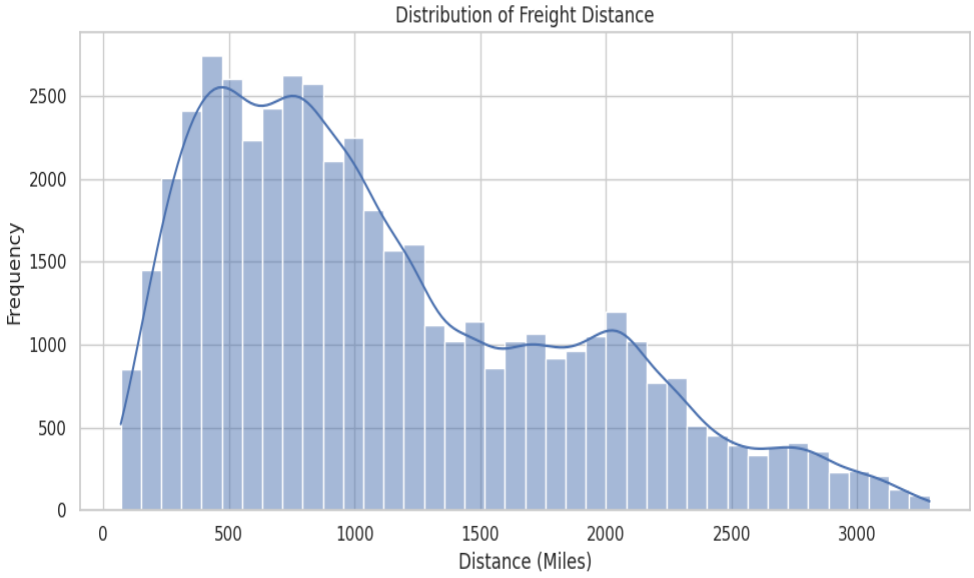
A. **Univariate analysis**:

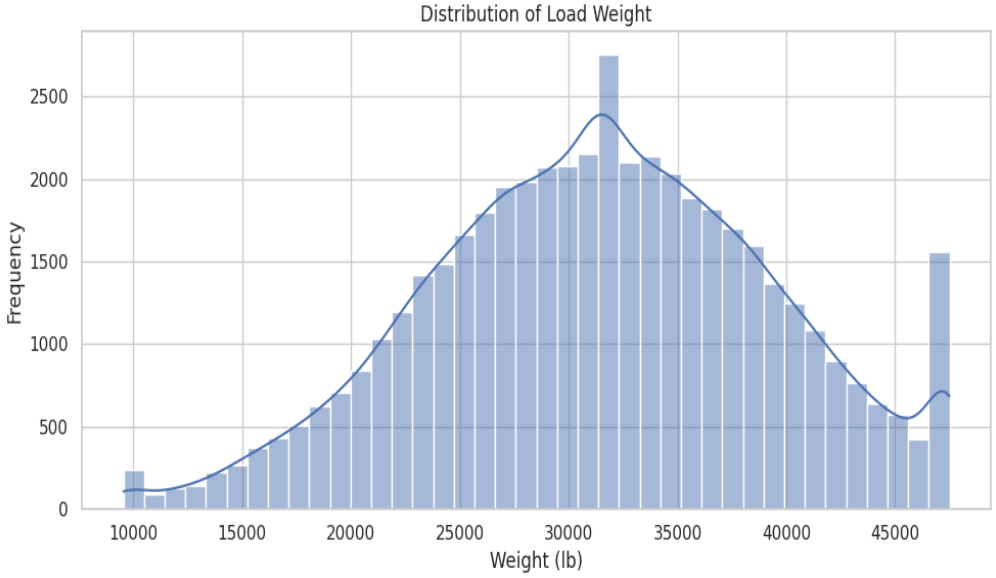


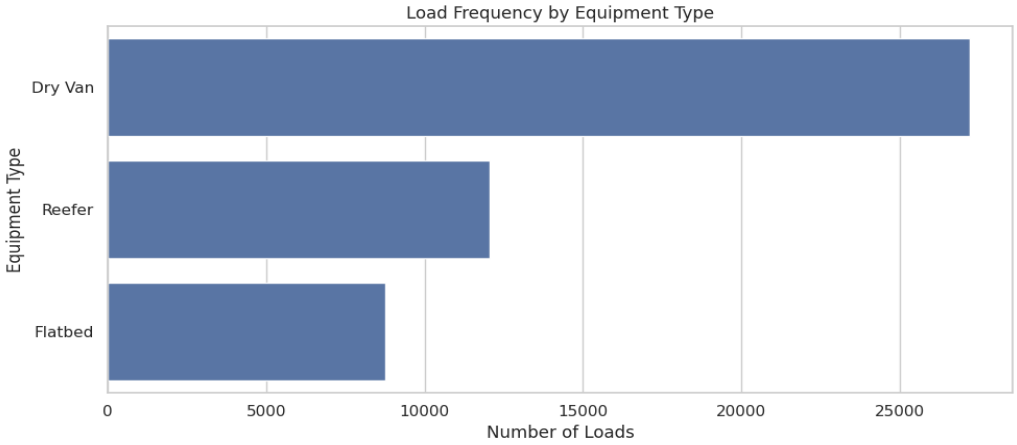
Freight rate is right skewed.



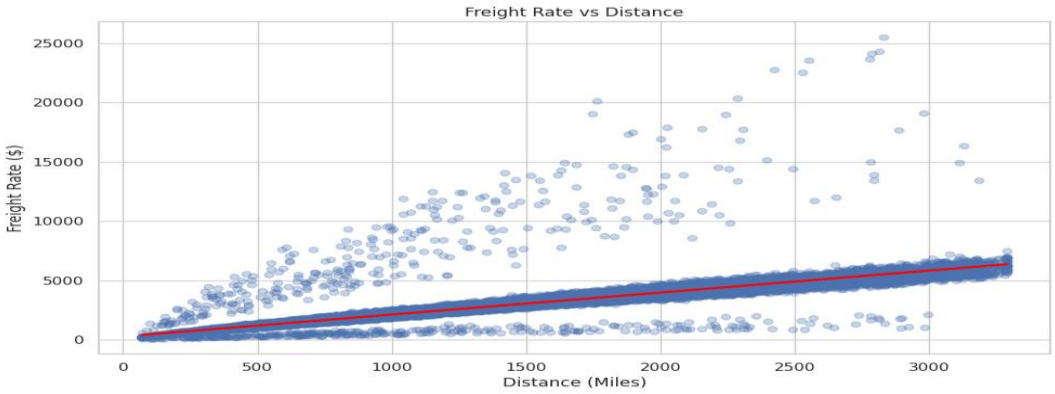
Freight rate has outliers.



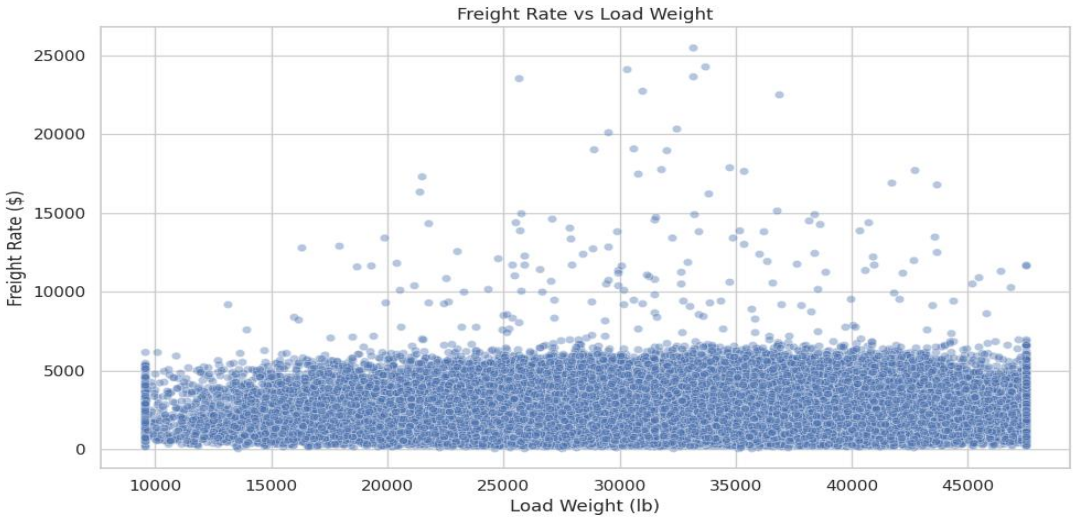




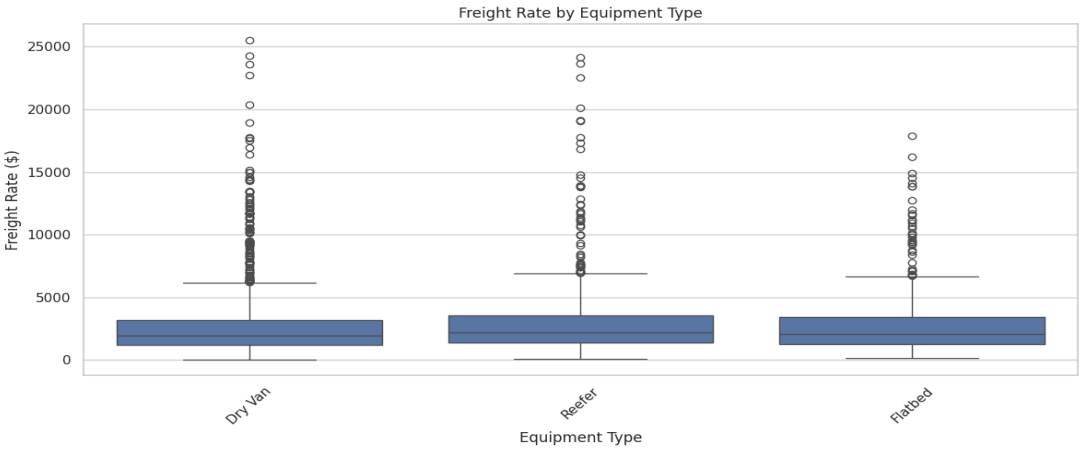
B. **Bivariate analysis**:



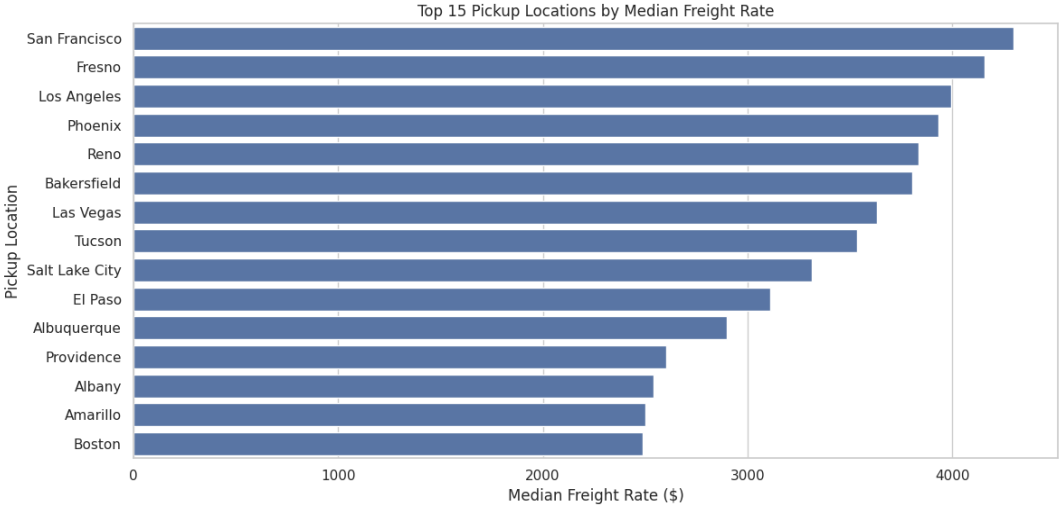
As distance increases freight rate increases.

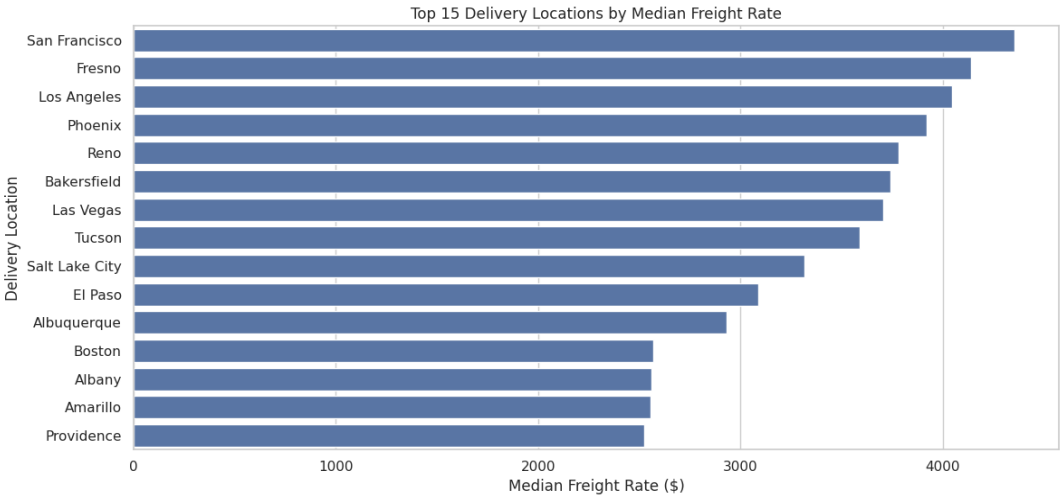


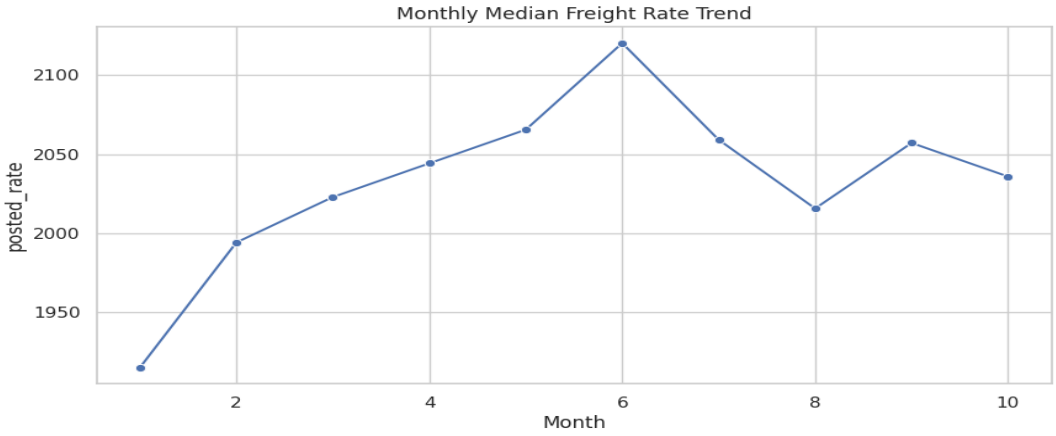
Higher load weights have higher freight rate.



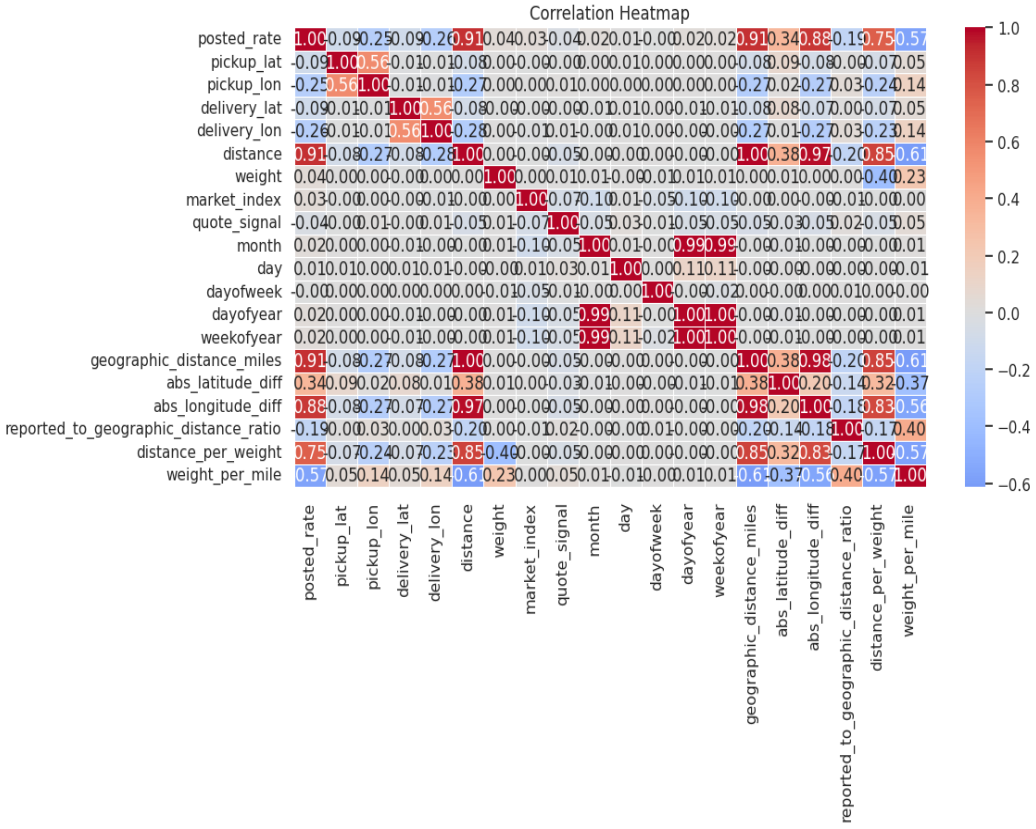
Every equipment has a different freight rate.







Month of June has higher freight rate.



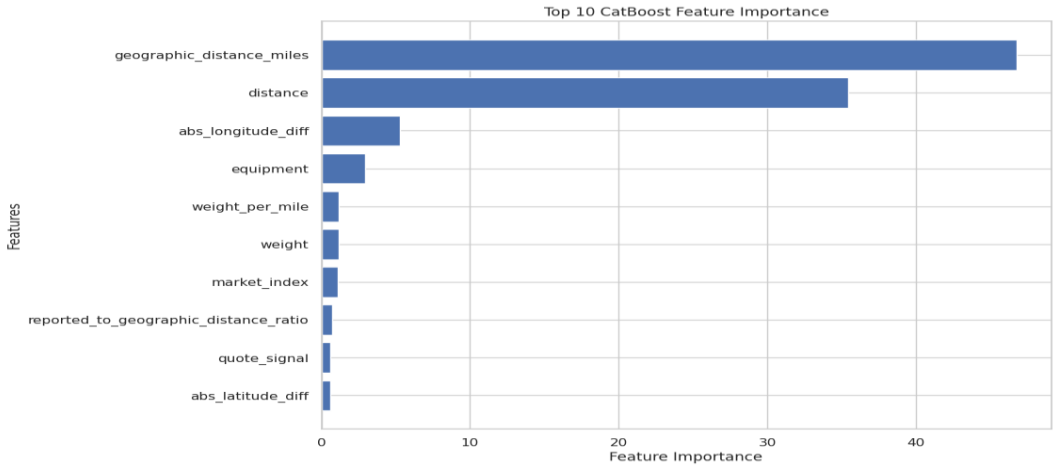
Some variables are strongly correlated with each other.

4. **Model Building**:

   Since CatBoostRegressor handles categorical data and non-linearity in the data well, it was selected for model building.

5. **Final prediction generation**:

   When tested onto the test data, it achieved a Root mean squared error (RMSE) of 0.1536 and Co-efficient of determination (R2 score) of 0.94 which indicated that the model is a good fit. Hence the freight rates for validation data were generated using this model. Also the important features identified by the model are as shown below.



6. **Final December Prediction**:

   The fixed December scenario contains 31 rows. The completed December prediction file was accepted by score.py, which reported: “Validated 31 fixed December predictions.”

